DESIGN MANAGEMENT

DESIGN THINKING IN EDUCATION: CAN WE IMPLEMENT DESIGN THINKING INTO MANAGEMENT SCHOOLS AND THE BUSINESS WORLD?

Gabrielle Conroy ID: 33893829 April 2017

Introduction

When it comes to education, one may be tempted to draw a line down the middle, separating and defining one side being the 'academic' and the 'logical', while the other being the 'creative'. Perhaps suggesting that educational practices are either traditional or progressive. The concept of potentially bringing these two worlds together, in this case the world of business and management, and that of design, is an interesting notion and will be the subject of this report. Within the question: Can we implement design thinking into management schools and the business world?, the ideas that will be explored are that of defining Design Thinking and Design Management, the teaching practices in both Design and Management schools, as well as how Design Thinking can be integrated into management education and the reasons for doing this. Drawing on relevant literature, theories, and the discussion of critics' views support the argument, providing some backdrop to the issue in question. It is considered within the paper, the possibility and potential of implementing Design Thinking not only into management schools and their curriculums, but into the business world in broader terms, allowing for the idea of this integration to be explored beyond the classroom setting. By doing so, one could argue that the teaching of business and management will be revolutionised, and will no longer cling to tradition as it once did, but embrace change in favour of a more progressive style of education.

Part One: What is Design thinking and Design Management?

Steve Jobs once said that "most people make the mistake of thinking design is what it looks like…Design is how it works". Though indeed a large part of design is made up from aesthetics, Design Thinking delves into the designer's mind and what processes are undertaken in order to come up with finished ideas which may produce products or lead onto improving services or even management. Though interlinked, one mustn't make the mistake of assuming that 'Design Thinking' and 'Design Management' are ultimately synonyms for one another - rather, that Design Thinking is an *element* of Design Management.

According to the 'design management institute', "Simply put, design management is the business side of design" (Dmi.org, 2017). It can be described as a business discipline, one that "encompasses the ongoing processes, business decisions, and strategies that enable innovation" (Dmi.org, 2017). Now, for want of a definition of Design Thinking, a methodology found under the umbrella of Design Management, IDEO CEO Tim Brown offers some insight:

> "Design Thinking…uses the designer's sensibility and methods to match people's needs with what is technologically feasible and what a viable business strategy can convert into customer value and market opportunity" (Naiman, 2017).

Perhaps an important thing to remember, as Linda Naiman, founder of 'Creativity at Work' states, "a design mindset is not problem-focused, it's solution focused" (Naiman, 2017).

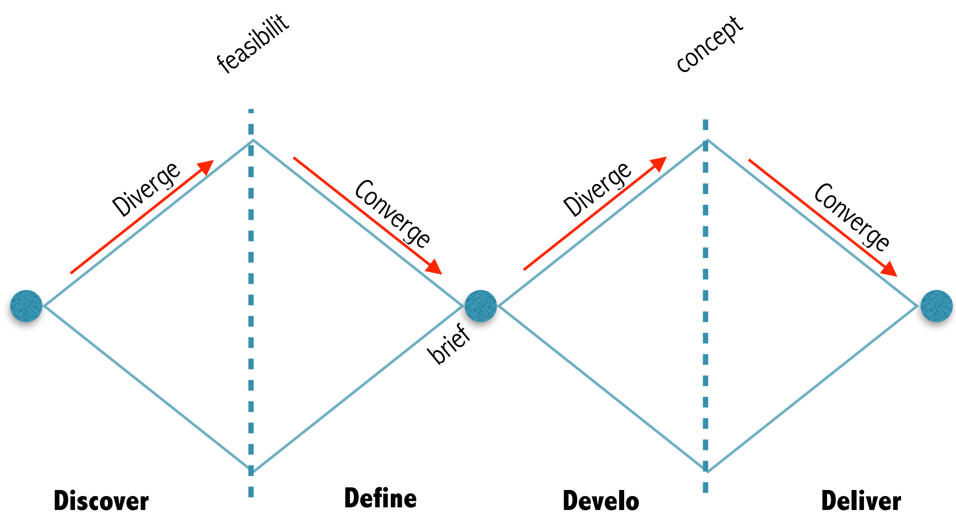
A valuable source to consider offers a definition of Design Thinking from the Stanford Graduate School of Business: *Design Thinking for Social Innovation - Stanford Social Innovation Review*. According to Brown & Wyatt, "as an approach, Design Thinking taps into capacities we all have but that are overlooked by more conventional problem-solving practices. Not only does it focus on creating products and services that are human centred, but the process itself is also deeply human" (Brown and Wyatt, 2010).

Bearing in mind that Design Thinking is commonly used to find solutions for clients and aids businesses, this addresses the question of: Can design and management be connected? Can Design Thinking aid business management? And if so, should we then be teaching Design Thinking in business schools? This report will examine this question as well as evaluate the differences between teaching in management schools and business schools, and analyse how Design Thinking can help the creative process and if it can create a competitive advantage.

Part Two: Teaching and education in Design Schools

It is assumed that in Design schools, students are taught how to be "creative". However, some would argue that one cannot teach creativity, it is not a discipline, but an innate ability. The very word could be subjective, as it begs the question: What is creativity? Established schools of Design Thinking, such as the first two to be founded: the d.school at Stanford University and the d.school of the Hasso-Plattner Institute, claim to teach their students how to work together to "solve big problems in a human centred way". It seems as though through Design Thinking, students are actively encouraged to 'free their mind' as it were and consider a whole range of possibilities, a teaching style which one may not find in other disciplines such as business, where established models are used time and time again and adhered to by teaching staff and the like.

Countless times has the Design Council's 'double diamond' been brought up when discussing Design Thinking - it seems to be a key element in Design Thinking education. This loose framework model teaches students to discover, define, develop then deliver, going through a process of brainstorming ideas then narrowing them down multiple times.



One could argue that the teaching style of Design Thinking challenges students in a way different to other subject areas. Referring to *Design Thinking for Education: Concepts and Applications in Teaching and Learning* :

> "Several scholars and researchers take the perspective that design thinking is particularly suited to dealing with ill-defined or wicked problems that cannot be fully resolved in part because they cannot be fully comprehended" (Koh et al., 2015, pg.2)

Thus, students are taught problem-solving skills in Design education, how to solve "wicked problems" which are here defined as ones that "cannot be easily described or defined, and they can be changeable, shifting in nature overtime" (Koh et al., 2015, pg.2). This implies that Design Thinking, as a practice, and wider, Design in general, is able to keep up with the constant fluctuations present in the current world. That the practice teaches students to adapt to their situation.

This can be shown in contrast to what some may call "the traditional system" - business/ management schools may fall under this category - in which the system is "predicted on the assumption that knowledge is relatively stable and there is a core set of knowledge that should be understood by all" (Koh et al., 2015, pg.8). In the eyes of many, this could be one of the core differences between the two styles of education.

It could be feasible to suggest that, from an educational perspective, Design Thinking has the potential to "contribute to the development of the creative and adaptive capacities of students" (Koh et al., 2015, pg.8) more than business disciplines do. The idea that "Design Thinking thrives on ambiguity and uncertainty" (Koh et al., 2015, pg.12) leads us to believe that this style of teaching and teaching methods broaden student's ways of thinking. Perhaps then, as a discipline, design can lead as an example and can create awareness of potential problem areas that exist within the 'traditional' school system, or the limitations, then possibly revolutionise the way students are taught, and how they learn. In short, design thinking should be integrated with existing teaching methods and is the way forward in education.

Part Three: Educational Practice in Business/Management schools

Though far from being completely linked and integrated with one another, there is some evidence of the management world considering the potential of design. Business magazines such as Fortune and BusinessWeek regularly feature design successes in their issues and draw attention to the importance of design for managers.

As mentioned earlier, the traditional school system, which here refers to the likes of business and management schools, views knowledge as something which is steady and unchanging, which leaves little room for adaption or adjustment. A critical source which proves relevant to the discussion of the teaching in management schools and the potential of Design Thinking is: *Design Thinking and How It Will Change Management Education: An Interview and Discussion* (Dunne and Martin, 2006). Roger Martin, dean of the Rotman School of Management, comments on the weaknesses of business schools and their teaching style, arguing that integrating Design Thinking can help. In terms of teaching, author David Dunne states:

> "business schools are under intense criticism, and in the view of some, have reached a point of crisis. Both academics and management practitioners criticise MBA programs for their lack of relevance to practitioners, the values they impart to students, and their teaching methods" (Dunne and Martin, 2006, pg.512)

This is an interesting statement as it suggests that traditional education may no longer be 'the best' way to teach, thus challenging it and choosing to favour more progressive methods which attempt to 'keep up' with current changes and trends. Traditional referring to conventional, established methods of teaching which have been in place for several years, i.e. teaching the same information to students year, after year. Although being 'long-established' may warrant some merit, it does not necessarily mean that it is still relevant. The lack of integration and new information being taught on curriculums is also picked up on in *Shaping the Future of Business Education: Relevance, Rigor, and Life Preparation*: "most business schools today still teach primarily the traditional, long-established, functional courses with little or no integration" (Hardy and Everett, 2013, pg.30). Particularly when it comes to problem-solving techniques, it is apparent that within business education, students are confined in their thinking style as problems are presented to them in their respective groups according to function. With this in front of them, students must consider the problems through a narrow lense, they must adhere to the methods they have been taught on that specific course. As a result of this, students develop 'one-dimensional' approaches to solving problems which are complex and in-depth (Hardy and Everett, 2013).

To assume that the economic climate, business world, or the state of affairs in general is in the same situation as say ten years ago may seem foolish to those in favour of more modern, progressive education, such as Design Thinking advocates.

In *Shaping the Future of Business Education: Relevance, Rigor and Life Preparation,* author Hardy seems to agree with Martin, stating that "The need for tomorrow's managers to have a broader education is being acknowledged by a few business schools that have launched initiatives to integrate liberal arts into the business curriculum" (Hardy and Everett, 2013, pg.29). For example, at Lancaster University, undergraduates are given the option to double major, one major being in the business discipline Marketing, and the other in an arts subject - Design.

An apparent difference between the teaching style found within management schools and design schools is that in the former, deductive and inductive logic are generally used (the logic of what should be or what is), whereas designers favour using abductive logic (the logic of what might be) (Dunne and Martin, 2006). Martin seems to hold the opinion that management education, in order to improve now and in the future, needs to take note of the importance of design and 'become' more like the education of design.

> "today's business people don't need to understand designers better, they need to become designers" - Roger Martin

In terms of discovering people's needs, traditional and conventional data gathering methods (e.g. questionnaires, focus groups etc.) which are often used on business and management courses, may provide valuable insight however they do not delve deeper as the Design Thinking approach would. Simply asking consumers what they want does not compare to observing their behaviour, the latter tends to lead to quantum leaps. In short this could boil down to the opinionated statement: 'Customers do not know what they want'. There are examples of projects that have been successful with figureheads who have chosen to opt for a heavily intuition-lead approach. Take Henry Ford for example, who was famously quoted saying: "If I'd asked my customers what they wanted, they'd have said 'a faster horse'". Another being Steve Jobs, who was well-known for choosing to steer clear of making decisions based on market research, instead relying on his intuition:

> "A lot of times, people don't know what they want until you show it to them" - Steve Jobs

However, although Steve Jobs and the like may have been hugely successful, some may label such a strategy as "try at your own risk".

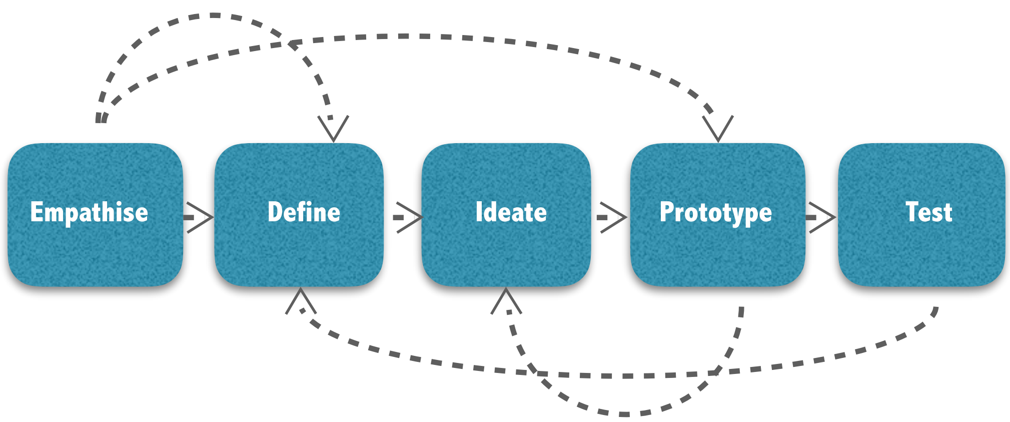
Intuition seems to be the driving force behind the process of Design Thinking, however, one should not assume that an organisation can be run on this alone. Again, on the other hand, being entirely dependent on rational and analytical thoughts, and data gathered from market research, can be a risky strategy. Therefore, using a combination of both, i.e. Design Thinking, can be the solution (Brown and Wyatt, 2010). Although described as a process, it is more often than not, non-sequential. This provides a contrast to the 'business-way' of thinking, which takes a more linear approach and is concerned with milestones - organisations typically follow this route rather than branching out into more progressive ways of working (Brown and Wyatt, 2010).

Part Four: How Design Thinking can be integrated into business education

In terms of how Design Thinking can be integrated into business education, Martin suggests that management schools need to teach its students to use abductive reasoning in combination with their existing style of using inductive and deductive reasoning (Dunne and Martin, 2006). Faculty needs to "be less doctrinaire in accepting only deductive and inductive logic and teaching in a way that suggests they are right and everyone else is wrong" (Dunne and Martin, 2006, pg.516).

In addition to this, Martin believes that MBA's need to learn how to listen to others and improve their collaboration skills, they need to become more open-minded to the reasoning process of others, as designers do. By moving away from the traditional MBA approach based on the Harvard model, business education can improve and adjust to become more progressive, thus preparing its students with more relevant skills to apply in the working world beyond university (Dunne and Martin, 2006).

Another difference between design and management education in schools highlighted by Martin, is the difference between the level of depth students go into when attempting to understanding what the consumer wants. Design students are taught how to 'empathise' with the user, it is the first stage in the Design Thinking process. The empathise stage involves activities such as "consulting experts to find out more about the area of concern through observing, engaging and empathising with people to understand their experiences and motivations, as well as immersing yourself in the physical environment to have a deeper understanding of the issues involved. Empathy is a crucial to a human-centred design process… [it] allows design thinkers to set aside his or her own assumptions about the world in order to gain insight into users and their needs" (Dam and Siang, 2017).



Also concerning consumers, Martin claims that MBA students in business schools seem to pay more attention to shareholders' interests than they do to customers' and society's. Martin's solution to this is to introduce integrative thinking into management education as this is a strategy that designers use. It does not have to be choice between which group receives the most attention, it can be both (Dunne and Martin, 2006).

Another criticism raised by Martin, and an interesting discussion point is the idea that business research doesn't produce new ideas but design does. One could argue that this is a valid point given that on many business and management courses, theories and models are still being taught that have been around for decades. For example, on many Undergraduate Marketing major courses, certainly on the one provided at Lancaster University, models such as the Boston Consulting Group Matrix and the GE-McKinsey Matrix are still covered, despite originally being introduced in the late 1960s and early 1970s. Though they are 'traditional' and have been taught for several years, as mentioned before, this does not necessarily mean that they are still applicable in today's world. In support of the statement that design produces new ideas rather than business, and in contrast to more conventional ways of learning and thinking, design students, through Design Thinking are taught to "recognise patterns, to construct ideas that have emotional meaning as well as being functional, and to express ourselves in media other than words or symbols" (Brown and Wyatt, 2010). This progressive, 'modern' way of thinking may indeed create new ideas and show that there are alternative routes to clinging to tradition.

Part Five: Why implement Design Thinking into Management education? How does it help the creative process?

It has been acknowledged that Design Thinking could be the way forward for management education, but how could it be? What reasons are there that can justify how good Design Thinking is? The answers could be given the heading, as the title of an article from the Graduate School of Stanford Business states, 'How Design Thinking Improves the Creative Process'. As the article states, more and more organisations have taken note of the emergence of Design Thinking as a method to solving problems, it no longer exists as just something IDEO use. The article is comprised of an interview with Stefanos Zenios, a professor of operations, information and technology at Stanford GSB, who provides some reasons as to why Design Thinking can improve the creative process.

The first issue that is raised is that of brainstorming. Through Design Thinking courses, students are taught the importance of brainstorming and how it can generate the wildest ideas which can lead to potential breakthroughs. Brainstorming, as Zenios mentions, can lead to great ideas which solve problems in a 'unique' way. As a method, it is designed to bring people together, who may come from different backgrounds, and have difference experiences, this makes the session more fruitful and interesting as a whole host of ideas can be generated. Again, as seems to be the theme with Design Thinking, brainstorming is a collaborative approach. As mentioned earlier, when referring to the interview with Roger Martin, collaboration seems to a feature which could be incorporated more into management education to improve it.

Another aspect to Design Thinking which aids the creative process, and could thus be implemented onto business courses, is the idea of "putting yourself in your user's shoes" (Zenios, 2016). Zenios recommends making things less formal, that instead of conducting an interview, as would be a common approach that MBAs would take, rather make it a conversation. 'Immersion' and 'empathy' are two terms that appear to be noteworthy in the article, again it is highlighted that Design Thinking is an approach which deeply concerns the user. Not simply asking the user what they want, but immersing yourself in their environment, and really understanding their feelings. The empathy stage of Design Thinking goes one step further than simply observation, as Zenios believes, "when you observe users, you see what they actually go through. But when you actually do it, that's when you can really understand how difficult some thing may be, or understand the emotions that some challenges are creating" (Zenios, 2016).

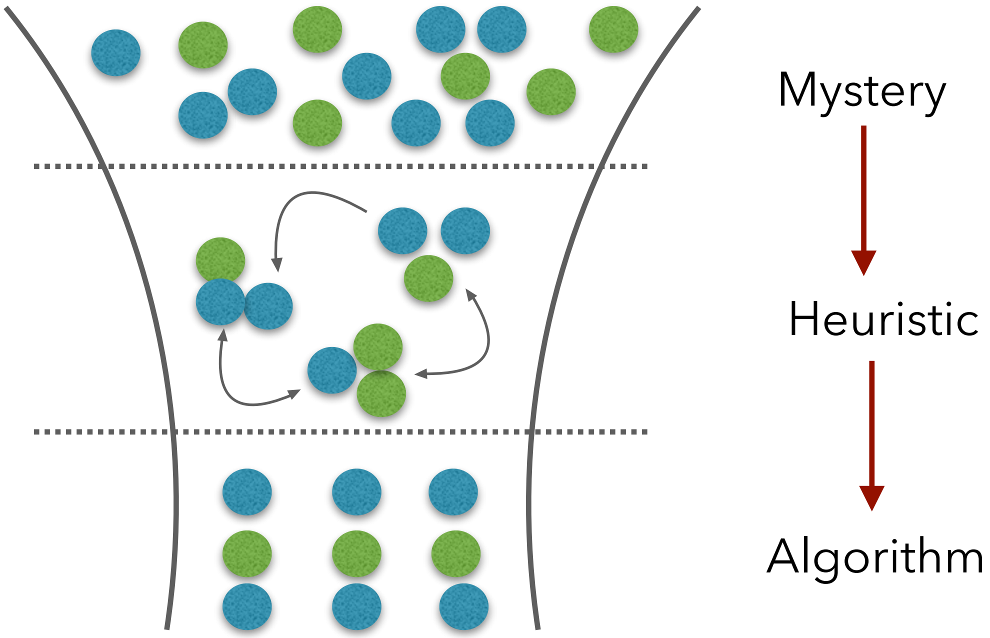
As for another way that Design Thinking can improve the creative process, the term 'prototype' emerges. By building a prototype this could improve the quality of feedback one would receive as consumers can physically interact with the initial product. This feedback could then lead to revaluation and reinvention, improvements could be made that will overall make a better product, thus the creative process has been improved. This also supports the fact that the Design Thinking process is not a sequential one, but this could be strength rather than a weakness. Potential problems areas that are discovered by users at the Prototype stage can take the designers back to the Ideate stage, who will then attempt to solve these problems. By looping back to stages, this allows designers to constantly make improvements before the final product is created. Therefore, management courses could 'take a leaf out design's book' as it were. By choosing not to cling to sequential and linear models or methods, the teaching of such subjects could encourage its students to become more creative thinkers, and to really appreciate the importance of customer feedback, not simply as something which is given after the fact, but is rather a part of the process.

Part Six: Why implement Design Thinking into companies? Can it create a competitive advantage?

With companies existing within heavily-saturated industries, striving to come up with a way to stand out from the crowd, for many Design Thinking can be the solution and the route to a competitive edge. There are existing examples of organisations which have chosen to take note of design methods and adopt these with regards to their own problems to solve (Dziersk, 2010). In Martin's book *The Design of Business,* he makes a case for the effectiveness of using Design Thinking as a business strategy, and part of the reason why Martin's case is so interesting and alluring is that Martin himself comes from the 'management world', not the 'design world' (Dziersk, 2010). Though he argues, why can't the two come together?

Martin outlines his opinions on where business strategy and thinking are flawed, and how adjusting these in accordance with Design Thinking methods can produce better results. A way that companies could create competitive advantage, according to Martin, is to move away from 'reliability bias', which is the groundwork for most businesses, and instead consider validity as equally important as reliability (Dziersk, 2010). Another point of interest is a theoretical construct created by Martin that he termed 'the Knowledge Funnel'. Martin describes the process as one which leading-businesses use in order to continue becoming more innovative and successful. However, in order for many businesses to improve, their understanding of the heuristic stage needs developing badly, as this is the stage that is generally problematic. Improvement of this could lead to greater innovation, and thus a competitive advantage among the market place.

The Knowledge Funnel:



Another reason as to why implementing Design Thinking can create a competitive advantage, is that it requires “less training efforts, less support, and also increased process excellence” (Voice, 2014). Thus, by using Design Thinking as a business strategy, firms can start to run more efficiently and effectively as less effort need be placed on training, service and customer research etc. (Voice, 2014). Companies who decide to follow the Design Thinking route could prove to work better and generate better results - beyond just functioning better in a timely fashion, design can also “lead to bold ideas of how to re-structure the workplace and inspire the workforce, unleashing human potential and unlocking opportunities for innovation” (Voice, 2014)

Conclusion

In conclusion, one could argue that it would be invaluable to implement Design Thinking into management education and the business world. Given that the discipline teaches them alternate, effective problem-solving skills, as well as adequately preparing them for the ‘outside world’ by teaching adaptivity in tandem with the changing climate around them, management education could surely benefit from incorporating Design Thinking as a business strategy. Rather than clinging to tradition, as being long-established does not necessarily merit relevance, business schools could embrace more progressive teaching methods which may contribute to the development of their students' creativity. This is not to say that the existing teaching methods found within business schools should be abandoned, but rather that it is integrated with that of the Design Thinking methods. Choosing to do this may result in management students becoming more open-minded in their thinking style as they will be taught to move away from the confines of a narrow perspective.

There already being evidence of the management world considering the potential of design, (Forbes, BusinessWeek and the like taking note), this only strengthens the argument that the two worlds should come together. Referring to relevant literature to look for some insight, several sources comment on the potential of the relationship, and do tend to favour Design's progressive nature. As author David Dunne commented, business schools are being criticised and their teaching methods seem to be lacking at this particular time. Incorporating methodologies taken from Design Thinking such as that of adductive logic , and seeking to delver deeper into customers needs by observation, and applying them to management courses could potentially improve them drastically. Although organisations should not run on intuition alone, it is needed to a certain extent and Design could teach Business this lesson. Creating a harmony of intuition and rational thoughts could be the key to business success.

When looking at Design frameworks and theories, examples such as the Design Thinking Process and the Double Diamond prove to offer some insight into how Designers work. As Roger Martin comments, Design students are taught to empathise, a skill that allows them to fully relate to consumers, this could be a skill that business students are be taught, improving their understanding of what their customers want., thus being able to provide better for them. An interesting idea that was discussed when researching literature, is that design produces new ideas whereas business research doesn't. One point supporting this statement being that it is common on business courses for models or theories that are decades old to still be taught.

Design Thinking is not only something that should be introduced onto curriculums in management schools, but also applied to companies and organisations who could also benefit from this approach. It is discussed in the report how Design Thinking can help the creative process. Using techniques such as brainstorming, prototyping and delving deeper into understanding what consumers want by using empathy methods can help firms improve their products/services and in turn produce better results.

The notion that Design Thinking can create a competitive advantage is also commented upon. One could argue that using Design Thinking can help an organisation to 'stand out from the crowd' as it were. By suggesting things such as considering validity as equally important as reliability bias, Roger Martin makes a case for the relationship of business and design being the way forward.

Taking this all into consideration, one could decide that Design Thinking does not have to be an approach that just designers use, it can be applied to other areas as well, because, as Arne Van Oosterom was once quoted saying: "Design Thinking is the glue between all disciplines".

References

Brown, T. and Wyatt, J. (2010). Design Thinking for Social Innovation. *Stanford Social Innovation Review*, 12(1).

Dam, R. and Siang, T. (2017). *5 Stages in the Design Thinking Process*. [online] The Interaction Design Foundation. Available at: https://www.interaction-design.org/literature/article/5-stages-in-the-design-thinking-process [Accessed 25 Apr. 2017].

Dmi.org. (2017). *What is Design Management? - Design Management Institute*. [online] Available at: http://www.dmi.org/?What_is_Design_Manag [Accessed 25 Apr. 2017].

Dunne, D. and Martin, R. (2006). *Design Thinking and How It Will Change Management Education: An Interview and Discussion*. 1st ed. [ebook] Academy of Management Learning & Education. Available at: http://citeseerx.ist.psu.edu/viewdoc/download?doi=10.1.1.193.9908&rep=rep1&type=pdf [Accessed 25 Apr. 2017].

Dziersk, M. (2010). Why Design Thinking Is the Next Competitive Advantage. [Blog] *Fast Company*. Available at: https://www.fastcompany.com/1628107/why-design-thinking-next-competitive-advantage [Accessed 25 Apr. 2017].

Hardy, G. and Everett, D. (2013). *Shaping the future of business education*. 1st ed. New York, NY: Palgrave Macmillan.

Koh, J., Chai, C., Wong, B. and Hong, H. (2015). *Design Thinking for Education: Concepts and Applications in Teaching and Learning*. 1st ed. Singapore: Springer.

Martin, R. (2009). *The design of business*. 1st ed. Boston, Mass.: Harvard Business Press.

Naiman, L. (2017). *Design Thinking as a Strategy for Innovation*. [online] Creativity at Work. Available at: http://www.creativityatwork.com/design-thinking-strategy-for-innovation/ [Accessed 25 Apr. 2017].

Voice, S. (2014). How Design Thinking Drives Competitive Advantage. [Blog] *Forbes*. Available at: https://www.forbes.com/sites/sap/2014/08/11/how-design-thinking-drives-competitive-advantage/#a414107504ac [Accessed 25 Apr. 2017].

Zenios, S. (2016). How design thinking improves the creative process: How entrepreneurs can take a big idea to a real-world product. [Blog] *Forbes India*. Available at: http://www.forbesindia.com/article/stanford/how-design-thinking-improves-the-creative-process/43071/1 [Accessed 25 Apr. 2017].